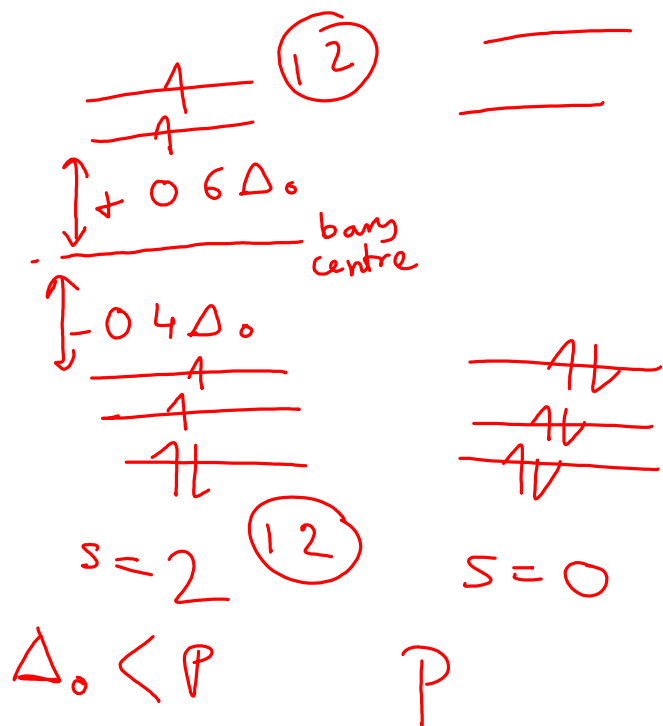


Practice Session 6

09-11-2024

1. Calculate the LFSE for high-spin and low-spin d^6 configuration in octahedral geometry.



CFSE/LFSE (hs)

$$= \boxed{(-0.4 \times 4)} \Delta_o + \cancel{P} + \boxed{(+0.6 \times 2)} \Delta_o$$

$$\downarrow s = (-0.4 \times 6) \Delta_o + 2P$$

$$= -2.4 \Delta_o + 2P \quad \Bigg| = 2.4 \Delta_o \cancel{*} 2P$$

2. Calculate the LFSE for d^7 configuration in tetrahedral geometry.

Tetrahedral complexes are high spin

$$\begin{array}{c} \begin{array}{c} (x^0 \ 4 \ \Delta x) \\ \hline \hline 1 \\ \hline \hline 1 \end{array} + 2 \\ \downarrow A_t \\ \begin{array}{c} 12 \\ \hline \hline 1V \end{array} \\ \downarrow \\ \begin{array}{c} E \\ (-0 \ 6 \ \Delta t) \end{array} \end{array}$$

$$\begin{aligned} \text{CFS E} &= (-0.6 \times 4) \Delta_T + (+0.4 \times 3) \Delta_t \\ &= -2.4 \Delta_T + 1.2 \Delta_t \\ &= \underline{\underline{-1.2 \Delta_T}} \end{aligned}$$

3. The magnetic moment of the complex $[\text{Mn}(\text{NCS})_6]^{4-}$ is $6.06\mu_B$. What is its electronic configuration?

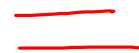
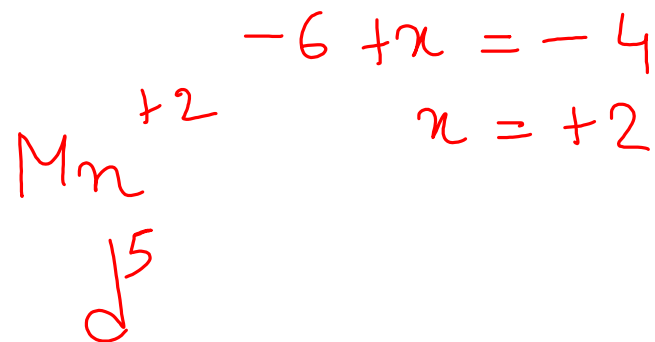
$n = \text{no. of unpaired } e^-$

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

$$= \sqrt{5(7)}$$

$$= \sqrt{35}$$

$$\approx 5.8$$

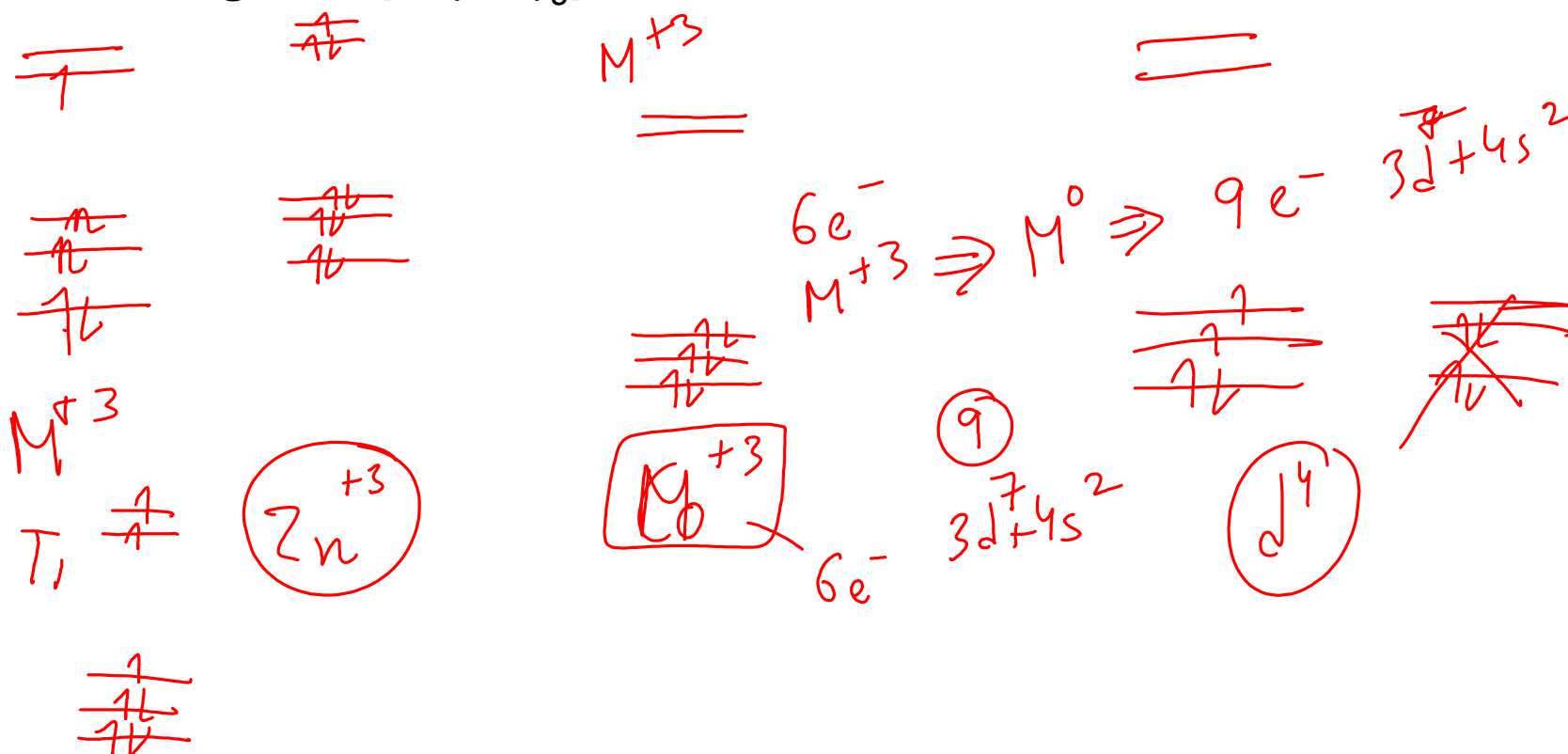


4. Predict the number of unpaired electrons for each of the following:

- A tetrahedral d^6 ion
- $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$

weak field ligand $\Rightarrow$ high-spin complex

- b. Diamagnetic $[\text{M}(\text{CN})_6]^{3-}$

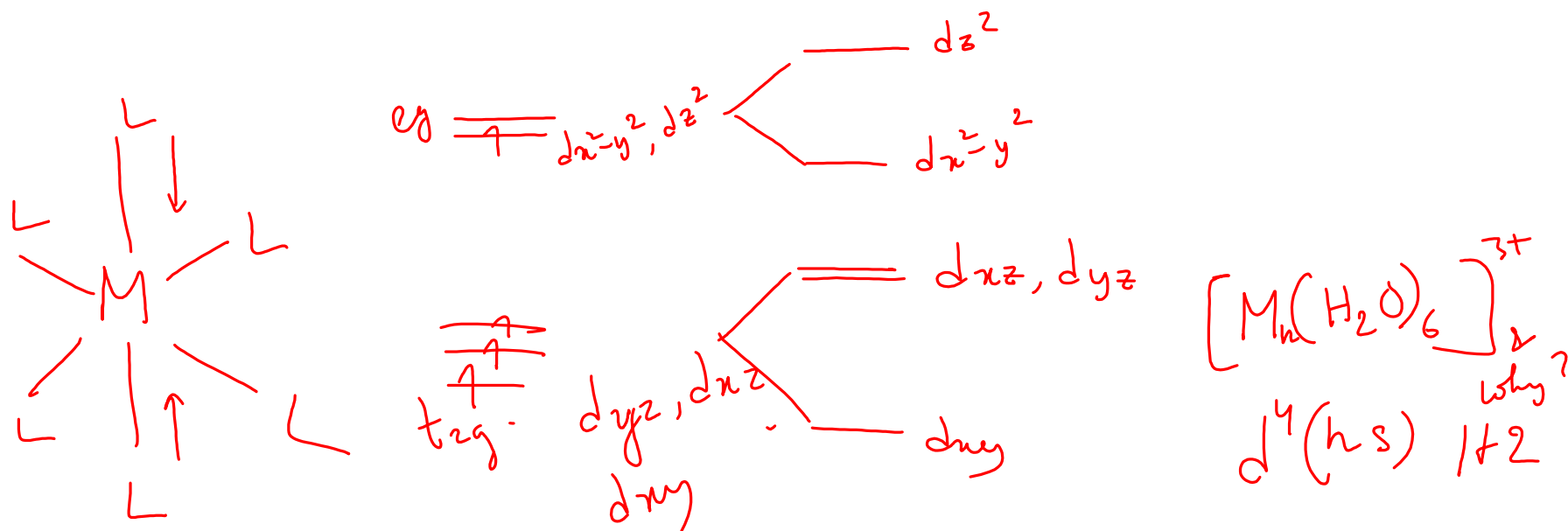


7. Identify the most likely transition metal **M**:

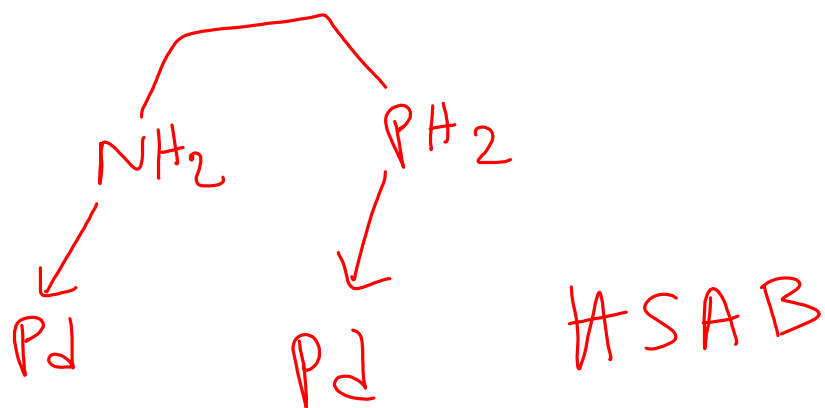
- a. $[\mathbf{M}(\text{H}_2\text{O})_6]^{3+}$, in which **M** is a second row transition metal and $\text{LFSE} = -2.4\Delta_0$
- b. Tetrahedral $[\mathbf{MCl}_4]^-$, which has five unpaired electrons and first-row transition metal **M**.

7. a. Explain the effect on d-orbital energies when an octahedral complex is compressed along the z axis.

7.b. Explain the effect on d-orbital energies when an octahedral complex is stretched along the z axis.

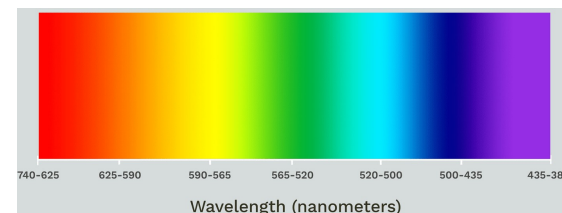
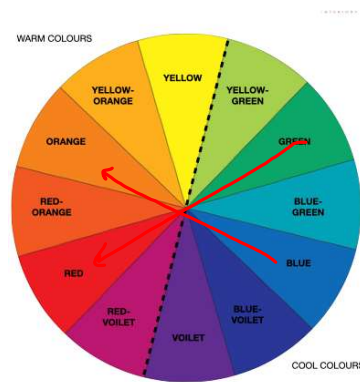


8. (2-amino-ethyl)phosphine ligand often acts as a bidentate ligand towards transition metals. When the ligand forms monodentate complexes with Pd, it bonds through its P atom. Suggest an explanation.

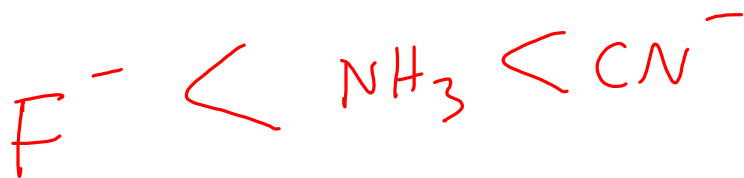


9. If one $[\text{CuL}]^{2+}$ solution is blue and another is green, which would be expected to have the higher values of Δ_o ?

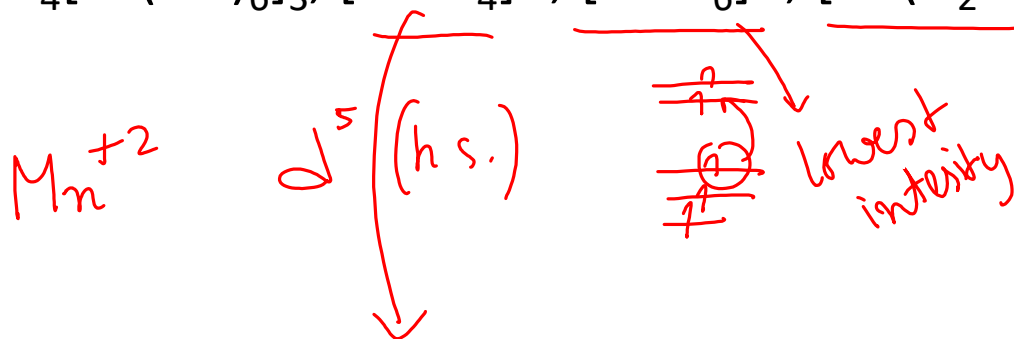
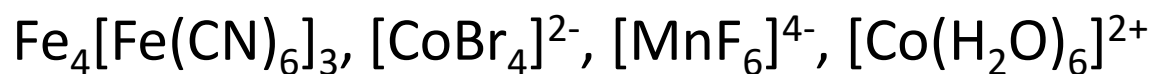
Color Blue $\rightarrow$ Absorbing Orange Δ_1
green $\rightarrow$ red Δ_2
 $\downarrow$
 $\lambda_{\text{red}} > \lambda_{\text{orange}}$
 $\Delta_1 > \Delta_2$



10. Consider the complex ions $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{CN})_6]^{3-}$, and $[\text{CoF}_6]^{3-}$. The wavelengths of absorbed electromagnetic radiation for these compounds (in no specific order) are 770 ^{F⁻} nm, 440 ^{NH₃} nm, and 290 ^{CN⁻} nm. Match the complex ion to the wavelength of absorbed electromagnetic radiation. Justify your matching with proper explanation.



- Arrange the given metal complexes in the increasing order of intensity of color (ϵ) shown by them. Justify your order by writing below each the status of the selection rules for these complexes :



- $[\text{Fe}(\text{CN})_6]^{3-}$ exhibits two sets of charge-transfer absorptions, one of lower intensity in the visible region of the spectrum, and one of higher intensity in the ultraviolet. $[\text{Fe}(\text{CN})_6]^{4-}$, however, shows only the high-intensity charge transfer in the ultraviolet. Explain.

- The isoelectronic ions VO_4^{3-} , CrO_4^{2-} , and MnO_4^- all have intense charge-transfer transitions. The wavelengths of these transitions increase in this series, with MnO_4^- having its charge-transfer absorption at the longest wavelength. Suggest a reason for this trend.

- The stepwise stability constants in aqueous solution at 25C for the formation of the ions $[\text{M}(\text{en})(\text{H}_2\text{O})_4]^{2+}$, $[\text{M}(\text{en})_2(\text{H}_2\text{O})_2]^{2+}$, and $[\text{M}(\text{en})_3]^{2+}$ for copper and nickel are given in the table. Why is there such a difference in the third values?

	$[\text{M}(\text{en})(\text{H}_2\text{O})_4]^{2+}$	$[\text{M}(\text{en})_2(\text{H}_2\text{O})_2]^{2+}$	$[\text{M}(\text{en})_3]^{2+}$
Cu	3×10^{10}	1×10^9	0.1 (estimated)
Ni	2×10^7	1×10^6	1×10^4